



UNIVERSIDAD TECNICA FEDERICO SANTA MARIA

Master's Thesis

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Thesis as a requirement to obtain the degree of
Magister en Ciencias de la Ingeniería Electrónica

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January 2025 , Valparaíso, Chile



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Acknowledgments

This is the abstract

Abstract

The global maritime industry is increasingly prioritizing sustainability, promoting the adoption of various hybrid and fully electric propulsion systems for different vessel types. This document presents a design and sizing methodology of an energy storage system for a hybrid tugboat, customized to meet the operational demands while minimizing environmental impact. The proposed framework integrates analysis of load profile, energy storage sizing, battery technologies selection, and propulsion system optimization to achieve an optimal balance between performance, and emissions reduction. The results demonstrate that a well-designed energy storage system for a hybrid tugboat can achieve reductions in fuel consumption and greenhouse gas emissions without compromising the availability and robustness required for towing and transit operations.

Resumen

La industria marítima mundial está dando cada vez más prioridad a la sostenibilidad, promoviendo la adopción de diversos sistemas de propulsión híbridos y totalmente eléctricos para diferentes tipos de buques. En este documento se presenta una metodología de diseño y dimensionamiento de un sistema de almacenamiento de energía para un remolcador híbrido, personalizado para satisfacer las exigencias operativas y minimizar el impacto medioambiental. El marco propuesto integra el análisis del perfil de carga, el dimensionamiento del almacenamiento de energía, la selección de tecnologías de baterías y la optimización del sistema de propulsión para lograr un equilibrio óptimo entre el rendimiento y la reducción de emisiones. Los resultados demuestran que un sistema de almacenamiento de energía bien diseñado para un remolcador híbrido puede lograr reducciones en el consumo de combustible y las emisiones de gases de efecto invernadero sin comprometer la disponibilidad y la robustez necesarias para las operaciones de remolque y tránsito.

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Chapter 1

Appendix

1.1 CO₂ tax emissions

In countries such as Finland, there is a so-called carbon tax, which seeks to mitigate the effects of climate change by discouraging the use of fossil fuels [88].

Based on this, a carbon tax of \$50/ton_{CO₂} will be assumed, and the impact this has on economic sensitivity variables will be determined, since this tax mainly affects the OPEX related to each battery.

1.1.1 Net Present Value (NPV)

Table 1.1: Calculated NPV, with carbon taxes, without subsidies (small bank).

Battery Bank 15% P _{nom}			
Lifespan	10 years	15 years	20 years
NPV _{NMC}	\$241,940	\$1,545,943	\$2,433,426
NPV _{LFP}	\$1,929,657	\$3,354,487	\$4,324,203
NPV _{LTO}	\$1,007,204	\$2,587,384	\$3,662,827

Table 1.2: Calculated NPV, with carbon taxes and without subsidies (large bank).

Battery Bank 25% P _{nom}			
Lifespan	10 years	15 years	20 years
NPV _{NMC}	-\$3,340,207	-\$2,156,302	-\$1,350,557
NPV _{LFP}	-\$71,650	\$1,384,116	\$2,374,886
NPV _{LTO}	-\$1,792,895	\$176,901	\$1,517,511

1.1.2 Internal Rate of Return (IRR)

Table 1.3: Calculated IRR, with carbon taxes and without subsidies (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR _{NMC}	9.18%	13.29%	14.69%
IRR _{LFP}	19.91%	22.67%	23.43%
IRR _{LTO}	12.52%	16.17%	17.34%

Table 1.4: Calculated IRR, with carbon tax and without subsidies (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR _{NMC}	-3.08%	3.01%	5.52%
IRR _{LFP}	7.70%	12.03%	13.54%
IRR _{LTO}	1.63%	6.91%	8.95%

1.1.3 Return on Investment (ROI)/ Payback Period

Table 1.5: Calculated period for positive ROI, with carbon taxes and without subsidies (small bank).

Battery Bank 15% P_{nom}	
	Period
ROI _{NMC} > 0	≈ 6 years and 5 months
ROI _{LFP} > 0	≈ 4 years and 3 months
ROI _{LTO} > 0	≈ 5 years and 7 months

Table 1.6: Calculated period for positive ROI, with carbon taxes and without subsidies (large bank).

Battery Bank 25% P_{nom}	
	Period
ROI _{NMC} > 0	≈ 11 years and 7 months
ROI _{LFP} > 0	≈ 6 years and 7 months
ROI _{LTO} > 0	≈ 8 years and 11 months

1.2 Government grants

With the aim of contributing to carbon neutrality in the global maritime transport sector, there are government programs that promote the use of clean technologies that accelerate decarbonization, such as the Ministry of Oceans and Fisheries of the Republic of Korea (MOF), which announced its "First National Plan for the Development and Popularization of Eco-friendly Ships (2021-2030)," known as the "2030-K Eco-friendly Ship Promotion Strategy" [89].

This strategy explores advanced zero-emission technologies that can reduce greenhouse gas (GHG) emissions in areas such as ship design, future fuels, renewable energy, and equipment, among others. In this way, the Ministry of Finance is facilitating the conversion of 15% of Korean-flagged ships (528 out of a total of 3,542), for both the public and private sectors.

This will be considered a 10% government subsidy, which will directly affect the CAPEX of each design and the economic sensitivity variables.

1.2.1 Net Present Value (NPV)

Table 1.7: Calculated NPV, without taxes and with government grants (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	-\$150,908	\$925,272	\$1,657,703
NPV_{LFP}	\$1,438,809	\$2,635,817	\$3,405,480
NPV_{LTO}	\$670,356	\$2,022,713	\$2,943,104

Table 1.8: Calculated NPV, without taxes and with government grants (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	-\$3,545,391	-\$2,261,373	-\$1,992,502
NPV_{LFP}	-\$497,334	\$698,545	\$1,512,441
NPV_{LTO}	-\$1,902,694	-\$201,222	\$956,771

1.2.2 Internal Rate of Return (IRR)

Table 1.9: Calculated IRR, without taxes and with Government grants (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	7.17%	11.57%	13.13%
IRR_{LFP}	18.04%	21.01%	21.86%
IRR_{LTO}	11.39%	15.16%	16.44%

Table 1.10: Calculated IRR, without taxes and with Government grants (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	-5.47%	1.06%	3.82%
IRR_{LFP}	5.66%	10.30%	11.98%
IRR_{LTO}	0.63%	6.08%	8.21%

1.2.3 Return on Investment (ROI)/ Payback Period

Table 1.11: Calculated period for positive ROI, without carbon taxes and with government grants (small bank).

Battery Bank 15% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 7 years
$ROI_{LFP} > 0$	≈ 4 years and 6 months
$ROI_{LTO} > 0$	≈ 5 years and 10 months

Table 1.12: Calculated period for positive ROI, without taxes and with Government grants (large bank).

Battery Bank 25% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 14 years and 9 months
$ROI_{LFP} > 0$	≈ 8 years
$ROI_{LTO} > 0$	≈ 10 years and 4 months

1.3 Taxes and Grants

The following section presents the effect of applying taxes and government subsidies. The values are as follows:

- Carbon Tax: \$50/tCO₂
- A government subsidy of 10% on the cost of materials purchased for the development of the hybrid propulsion system.

1.3.1 Net Present Value (NPV)

Table 1.13: Calculated NPV, with taxes and government grants (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	\$675,690	\$1,979,693	\$2,867,167
NPV_{LFP}	\$2,265,407	\$3,690,237	\$4,659,953
NPV_{LTO}	\$1,496,954	\$3,077,134	\$4,152,577

Table 1.14: Calculated NPV, with taxes and government grants (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	-\$2,602,457	-\$1,418,552	-\$612,807
NPV_{LFP}	-\$445,600	\$1,901,366	\$2,892,126
NPV_{LTO}	-\$929,145	\$1,040,651	\$2,381,261

1.3.2 Internal Rate of Return (IRR)

Table 1.15: Calculated IRR, with taxes and government grants (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	17.57%	15.34%	16.58%
IRR_{LFP}	23.25%	25.67%	26.28%
IRR_{LTO}	15.32%	18.61%	19.61%

Table 1.16: Calculated IRR, with taxes and government grants (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	-1.34%	4.45%	6.78%
IRR_{LFP}	10.00%	13.99%	15.34%
IRR_{LTO}	3.60%	8.56%	10.42%

1.3.3 Return on Investment (ROI)/ Payback Period

Table 1.17: Calculated period for positive ROI, with taxes and government grants (small bank).

Battery Bank 15% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 5 years and 10 months
$ROI_{LFP} > 0$	≈ 3 years and 10 months
$ROI_{LTO} > 0$	≈ 5 years

Table 1.18: Calculated period for positive ROI, with taxes and government grants (large bank).

Battery Bank 25% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 11 years and 6 months
$ROI_{LFP} > 0$	≈ 6 years and 7 months
$ROI_{LTO} > 0$	≈ 8 years and 11 months

1.4 Three Level Buck Boost Converter

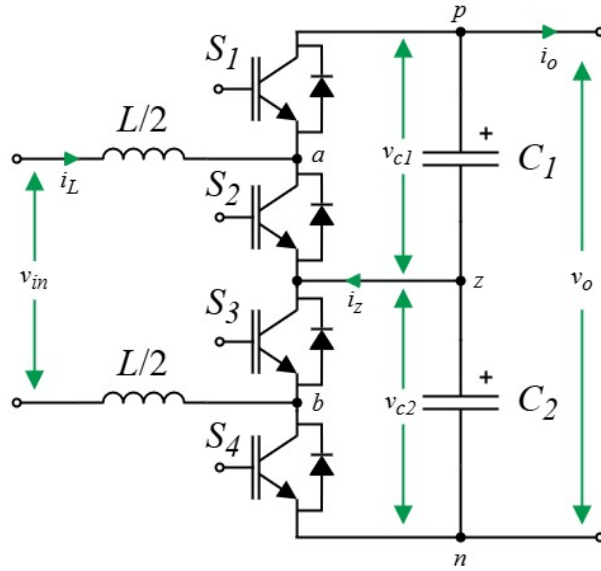


Figure 1.1: BusDC.

1.4.1 Modelling

It is assumed that C_1 and C_2 are equal.

$$C_1 = C_2 = C \quad (1.1)$$

State E0

$$\begin{aligned} v_{in}(t) - v_{L/2}(t) - v_{L/2}(t) &= 0 \\ v_{in}(t) - (L/2) \cdot \frac{di_L(t)}{dt} - (L/2) \cdot \frac{di_L(t)}{dt} &= 0 \\ \Rightarrow L \cdot \frac{di_L(t)}{dt} &= v_{in}(t) \end{aligned} \quad (1.2)$$

$$\begin{aligned} i_{c1}(t) &= -i_o(t) \\ \Rightarrow C \cdot \frac{dv_{c1}(t)}{dt} &= -i_o(t) \end{aligned} \quad (1.3)$$

$$\begin{aligned} i_{c2}(t) &= -i_o(t) \\ \Rightarrow C \cdot \frac{dv_{c2}(t)}{dt} &= -i_o(t) \end{aligned} \quad (1.4)$$

State E1

$$\begin{aligned}
v_{in}(t) - v_{L/2}(t) - v_{c1}(t) - v_{L/2}(t) &= 0 \\
v_{in}(t) - (L/2) \cdot \frac{di_L(t)}{dt} - v_{c1}(t) - (L/2) \cdot \frac{di_L(t)}{dt} &= 0 \\
\Rightarrow L \cdot \frac{di_L(t)}{dt} &= v_{in}(t) - v_{c1}(t)
\end{aligned} \tag{1.5}$$

$$\begin{aligned}
i_{c1}(t) &= i_L(t) - i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c1}(t)}{dt} &= i_L(t) - i_o(t)
\end{aligned} \tag{1.6}$$

$$\begin{aligned}
i_{c1}(t) &= i_L(t) + i_{c2}(t) \\
i_L(t) - i_o(t) &= i_L(t) + i_{c2}(t) \\
-i_o(t) &= i_{c2}(t) \\
\Rightarrow C \cdot \frac{dv_{c2}(t)}{dt} &= -i_o(t)
\end{aligned} \tag{1.7}$$

State E2

$$\begin{aligned}
v_{in}(t) - v_{L/2}(t) - v_{c2}(t) - v_{L/2}(t) &= 0 \\
v_{in}(t) - (L/2) \cdot \frac{di_L(t)}{dt} - v_{c2}(t) - (L/2) \cdot \frac{di_L(t)}{dt} &= 0 \\
\Rightarrow L \cdot \frac{di_L(t)}{dt} &= v_{in}(t) - v_{c2}(t)
\end{aligned} \tag{1.8}$$

$$\begin{aligned}
i_{c1}(t) &= -i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c1}(t)}{dt} &= -i_o(t)
\end{aligned} \tag{1.9}$$

$$\begin{aligned}
i_{c2}(t) &= i_L(t) - i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c2}(t)}{dt} &= i_L(t) - i_o(t)
\end{aligned} \tag{1.10}$$

State E3

$$\begin{aligned}
v_{in}(t) - v_{L/2}(t) - v_{c1}(t) - v_{c2}(t) - v_{L/2}(t) &= 0 \\
v_{in}(t) - (L/2) \cdot \frac{di_L(t)}{dt} - v_{c1}(t) - v_{c2}(t) - (L/2) \cdot \frac{di_L(t)}{dt} &= 0 \\
\Rightarrow L \cdot \frac{di_L(t)}{dt} &= v_{in}(t) - v_{c1}(t) - v_{c2}(t)
\end{aligned} \tag{1.11}$$

$$\begin{aligned}
i_{c1}(t) &= i_L(t) - i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c1}(t)}{dt} &= i_L(t) - i_o(t)
\end{aligned} \tag{1.12}$$

$$\begin{aligned}
i_{c2}(t) &= i_L(t) - i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c2}(t)}{dt} &= i_L(t) - i_o(t)
\end{aligned} \tag{1.13}$$

General differential Model

$$\frac{di_L(t)}{dt} = \frac{v_{in}(t) - v_{c1}(t) \cdot \mathbf{S}_1[t] - v_{c2}(t) \cdot \mathbf{S}_2[t]}{L} \tag{1.14}$$

$$\frac{dv_{c1}(t)}{dt} = \frac{i_L(t) \cdot \mathbf{S}_1[t] - i_o(t)}{C} \tag{1.15}$$

$$\frac{dv_{c2}(t)}{dt} = \frac{i_L(t) \cdot \mathbf{S}_2[t] - i_o(t)}{C} \tag{1.16}$$

$$\tag{1.17}$$

where

$$\mathbf{S}_1, \mathbf{S}_2 \in \{0, 1\} \tag{1.18}$$

An average model consider:

$$\begin{aligned}
d_i &= \frac{t_i^{ON}}{t_i^{ON} + t_i^{OFF}} \\
S_1[t] &\Rightarrow d_1(t) \\
S_2[t] &\Rightarrow d_2(t) \\
d_1, d_2 &\in (0, 1)
\end{aligned} \tag{1.19}$$

If the change of variables is performed:

$$\begin{aligned}
u_i(t) &= \frac{d_1(t) + d_2(t)}{2} \\
u_v(t) &= \frac{d_1(t) - d_2(t)}{2} \\
\Rightarrow u_i(t) + u_v(t) &= d_1(t) \\
\Rightarrow u_i(t) - u_v(t) &= d_2(t)
\end{aligned} \tag{1.20}$$

In this way:

$$\frac{di_L(t)}{dt} = \frac{v_{in}(t) - \mathbf{u}_i(t) \cdot (v_{c1}(t) + v_{c2}(t)) - \mathbf{u}_v(t) \cdot (v_{c1}(t) - v_{c2}(t))}{L} \quad (1.21)$$

$$\frac{dv_{c1}(t)}{dt} = \frac{i_L(t) \cdot (\mathbf{u}_i(t) + \mathbf{u}_v(t)) - i_o(t)}{C} \quad (1.22)$$

$$\frac{dv_{c2}(t)}{dt} = \frac{i_L(t) \cdot (\mathbf{u}_i(t) - \mathbf{u}_v(t)) - i_o(t)}{C} \quad (1.23)$$

1.5 Linearization

The state variables to be linearized are defined.

$$\begin{aligned} x(t) &= [i_L(t) \quad v_{c1}(t) \quad v_{c2}(t)]^T \\ u(t) &= [d_1(t) \quad d_2(t)]^T \\ y(t) &= [i_L(t) \quad (v_{c1}(t) - v_{c2}(t)) \quad v_o(t)]^T \end{aligned} \quad (1.24)$$

where $x(t)$ are the state variables, $u(t)$ are the inputs of the system and $y(t)$ are the output of the system.

To perform linearization, a signal corresponding to its equilibrium point plus a small variation will be taken into account.

$$\begin{aligned} x(t) &= \Delta x(t) + x_Q \\ u(t) &= \Delta u(t) + u_Q \end{aligned} \quad (1.25)$$

In this way, and applying a low-order Taylor series to linearize, the linearized differential equations are found.

$$\frac{d\Delta i_L(t)}{dt} = \frac{\Delta v_{in}(t) - \Delta v_{c1}(t) \cdot d_{1Q} - \Delta v_{c2}(t) \cdot d_{2Q} - \Delta d_1(t) \cdot v_{c1Q} - \Delta d_2(t) \cdot v_{c2Q}}{L} \quad (1.26)$$

$$\frac{d\Delta v_{c1}(t)}{dt} = \frac{\Delta i_L(t) \cdot d_{1Q} + \Delta d_1(t) \cdot i_{LQ} - \Delta i_o(t)}{C} \quad (1.27)$$

$$\frac{d\Delta v_{c2}(t)}{dt} = \frac{\Delta i_L(t) \cdot d_{2Q} + \Delta d_2(t) \cdot i_{LQ} - \Delta i_o(t)}{C} \quad (1.28)$$

In this way, the matrices of interest can be found in state variables within the linearized differential equations.

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \Delta i_L(t) \\ \Delta v_{c1}(t) \\ \Delta v_{c2}(t) \end{bmatrix} &= \begin{bmatrix} 0 & -d_{1Q}/L & -d_{2Q}/L \\ d_{1Q}/C & 0 & 0 \\ d_{2Q}/C & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta i_L(t) \\ \Delta v_{c1}(t) \\ \Delta v_{c2}(t) \end{bmatrix} + \begin{bmatrix} -v_{c1Q}/L & -v_{c2Q}/L \\ i_{LQ}/C & 0 \\ 0 & i_{LQ}/C \end{bmatrix} \begin{bmatrix} \Delta d_1(t) \\ \Delta d_2(t) \end{bmatrix} \\ &\quad + \begin{bmatrix} 1/L & 0 \\ 0 & -1/C \\ 0 & -1/C \end{bmatrix} \begin{bmatrix} \Delta v_{in}(t) \\ \Delta i_o(t) \end{bmatrix} \\ \begin{bmatrix} \Delta i_L(t) \\ \Delta v_{c1}(t) - \Delta v_{c2}(t) \\ \Delta v_o(t) \end{bmatrix} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \Delta i_L(t) \\ \Delta v_{c1}(t) \\ \Delta v_{c2}(t) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta d_1(t) \\ \Delta d_2(t) \end{bmatrix} \end{aligned} \quad (1.29)$$

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